P1 Nov 2009

1.

1. Age of universe ~ 14 billion years

$= 14 \times 10^9 \times 365 \times 24 \times 60 \times 60$

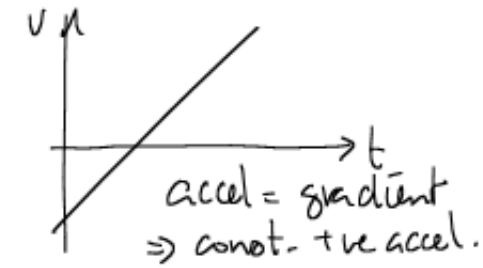
$\sim 4 \times 10^{17} \sim 10^{18}$ to nearest

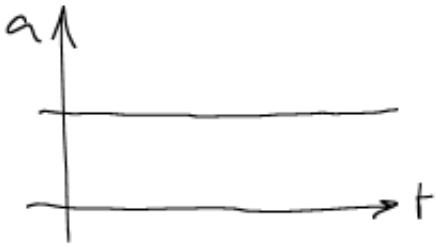
10^{18} doesn't seem old enough, I would have guessed something bigger.

B

2.

2.

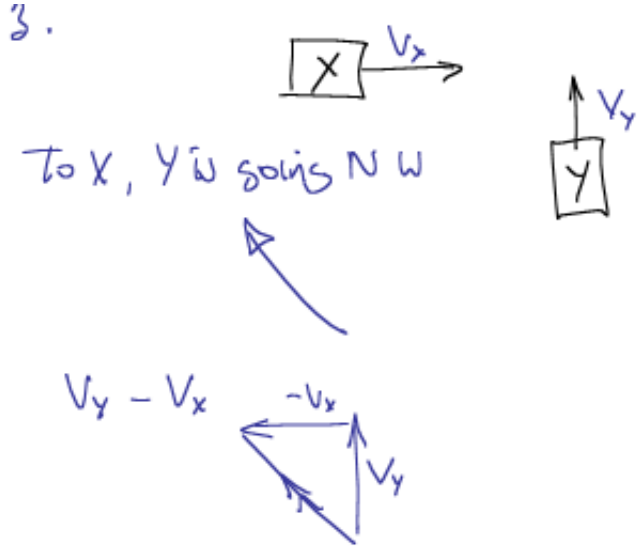




C

3.

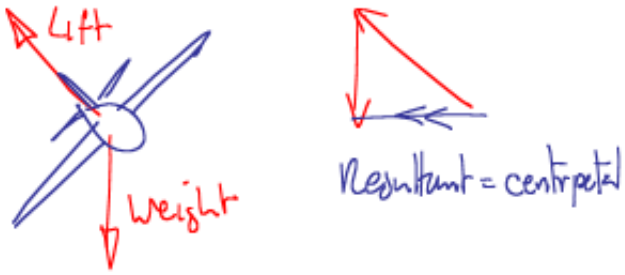
3.



D

4.

4.



Centripetal is not a 3rd force but the resultant of Lift and weight.

D

5.

5. A body will remain at rest or moving with const. velocity unless acted upon by an unbalanced force
⇒ body will not have constant velocity

A

6.

6. Uphill ⇒ increasing PE
constant speed ⇒ constant KE

Car also makes sound + heat
⇒ Chemical → PE + sound + heat

B

7.

7.

Potential – scalar
potential gradient = field strength = vector

A

8.

8.

The range is of course R but that's not an option

Horizontal comp = $u\cos\theta$

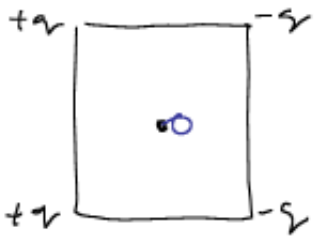


$R = u\cos\theta \times t$

B

9.

9.



Potential is a vector so the two +q cancel out the two −q

Potential = 0 A

10.

10. 6g of carbon is ½ mole

So will have twice as many atoms as ¼ mole of Oxygen

$\Rightarrow 4g$

C

A bit contrived, difficult for non native English speakers.

11.

11. $P = \frac{Q}{\Delta t} = mc\frac{\Delta\theta}{\Delta t}$

$60 = 0.1 \times c \times \frac{40}{100}$ $c = \frac{60\times 100}{40\times 0.1}$ A

12.

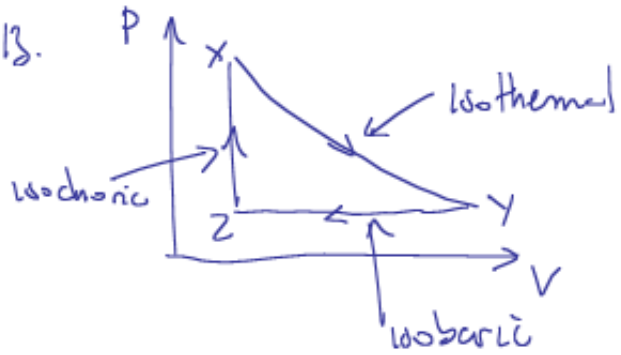
12. In real gases the forces between mols are not always negligible

B

Note that A and B are incompatible
If gas mols have PE then forces cannot be negligible
⇒ must be either A or B.

13.

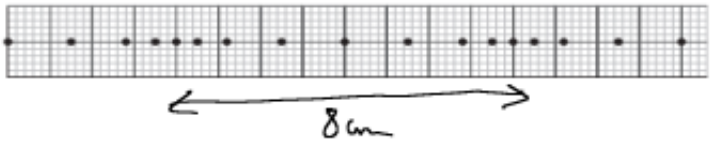
13.



A

14.

14.

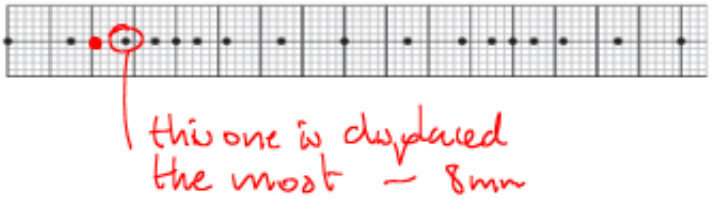


$\lambda = 8\,cm$

C

15.

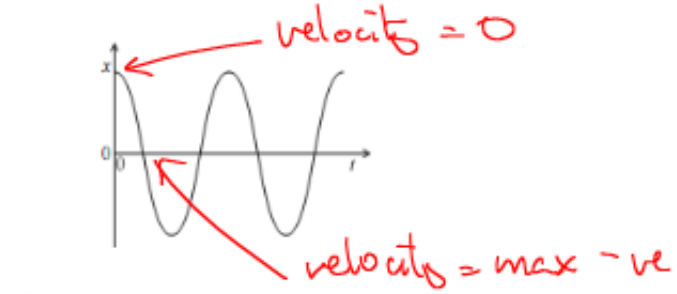
15. Amplitude = max displacement from original position

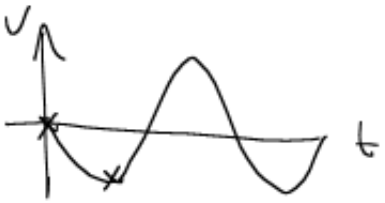


B

16.

16.





C

17.

17. Particle executes SHM ⇒ Total E is constant

D

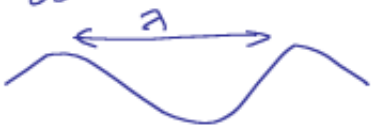
18.

18.

Standing wave – like a skipping rope
- different amplitudes
- same phase
- no energy transfer



Travelling wave – like a mexican wave
- same amplitude
- different phase
- energy transfer.



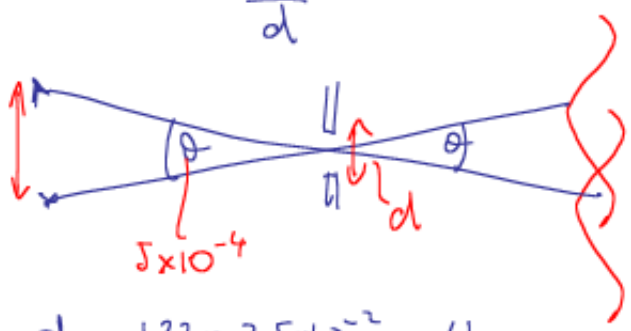
C

19.

19.

If just resolved then according to Rayleigh criteria

$\sin\theta = 1.22\frac{\lambda}{d} \simeq \theta = 5\times10^{-4}$

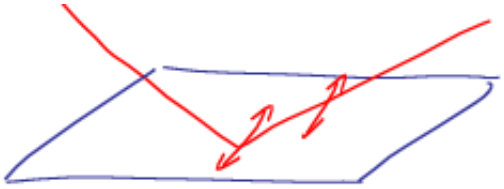


$d = \frac{1.22 \times 2.5\times10^{-2}}{5\times10^{-4}} = 61\ m$

A

20.

20.



Plane is parallel to surface

A

21.

21.

$W_{EF} = \Delta V q$

charge is positive so moves from high → low $\Rightarrow V_1 > V_2$

$W_{EF} = (V_1 - V_2) q$

C

22.

22.



$R = \frac{\rho l}{A} \Rightarrow A = \frac{\pi D^2}{4}$

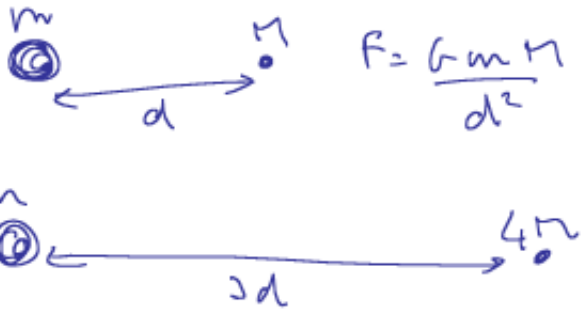
$\Rightarrow R = \frac{4\rho l}{\pi D^2}$

$R_2 = \frac{4 \times 2\rho \times 2l}{\pi (2D)^2} = \frac{4\rho l}{\pi D^2} = R_1$

B

23.

23.



$F = \frac{G m \cdot 4M}{(3d)^2} = \frac{4}{9}\frac{GmM}{d^2}$

$F = \frac{4}{9} \times 90 = 40N$

D

or $F \propto \frac{M}{d^2}$ $\frac{90}{F} = \frac{M}{d^2} \Big/ \frac{4M}{(3d)^2} = \frac{9}{4}$

$F = \frac{90}{9} \times 4$

24.

24.

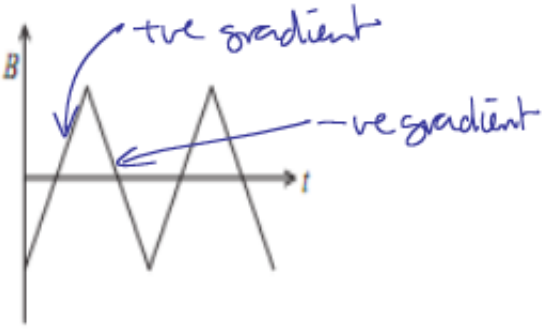
A test charge is a small charge that doesn't affect the field it is placed in. C is the best answer

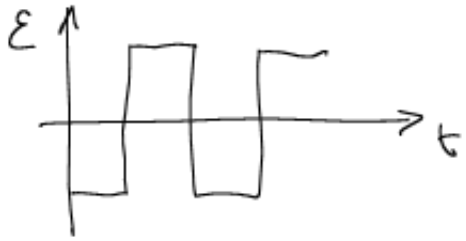
C

25.

25.

$\varepsilon = -\frac{d(BAN)}{dt}$





Note -ve sign

D

26.

26.

Power losses are due to

- Heating in coils
- Heating in core
- Hysteresis loss in core

Solid core – increases eddy currents ✗
large diameter – reduces resistance ✓
Low resistivity – reduces resistance ✓

C

27.

27. $A = Z + N$

Nucleon number = proton number + neutron number

$A = Z + N$

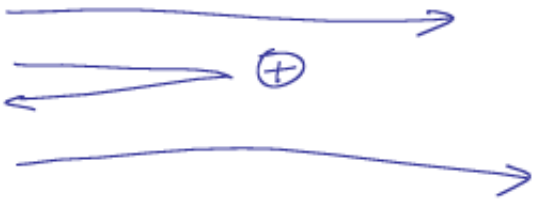
$\Rightarrow N = A - Z$

C

28.

28.

small +ve nucleus with large mass



B

29.

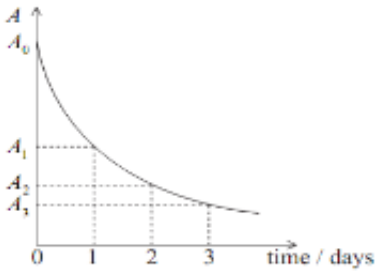
29.

Nucleus $\Rightarrow$ nuclear

D

30.

30.



half life = 1 day

$\Rightarrow \frac{A_0}{A_2} = 4$ same as $\frac{A_1}{A_3}$

A

Have to read this one carefully.
I. first assumed the question said $\frac{A_0}{A_1}$ but had to read again when got answer wrong.

31.

31. ψ^2 – probability of finding electron

C

32.

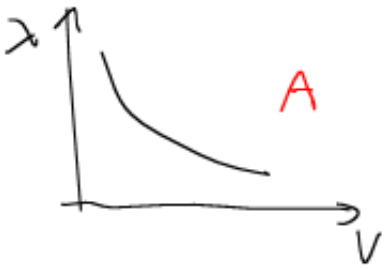
32.

$\lambda = \frac{h}{p} \Rightarrow \lambda \propto \frac{1}{mv}$

but $\frac{1}{2}mv^2 = V_q \Rightarrow v = \sqrt{\frac{2V_q}{m}}$

$mv = m\sqrt{\frac{2V_q}{m}} = \sqrt{2V_q m}$

$\lambda \propto \frac{1}{\sqrt{V}}$



33.

33.

- No time delay ✓
- light must be above threshold f
→ below max λ ✗
- not dependent on intensity ✗
- $E = hf - hf_0$ so does depend on f. ✗

A

Question actually says instantaneous which is not quite true.

34.

34

Winds are caused by air movement due to uneven heating by the sun

B

35.

35. $\text{Effic} = \frac{\text{Power out}}{\text{Power in}} \times 100$

$\Rightarrow \text{Power in} = \frac{1000}{20} \times 100 = 5000\text{MJ}$

in 1sec — 5000MJ

$\text{Energy density} = \frac{\text{Energy}}{\text{mass}}$

$\Rightarrow \text{Mass} = \frac{5000}{50} = 100\text{kg}$

D

36.

36.

$P = \sigma T^4$

$\frac{P_{Earth}}{P_{Venus}} = \frac{T_{Earth}^4}{T_{Venus}^4} = \left(\frac{300}{600}\right)^4$

$= 1/2^4 = 1/16$

D

37.

37. $\Delta V = V_0 \gamma \Delta T$

fractional increase $= \frac{\Delta V}{V_0} = \gamma \Delta T$

C

38.

38.

- SG power is max efficient ✓
- burning waste → CO_2 ✗
- wind power does not produce CO_2 ✓
- CO_2 capture — reduces CO_2 ✓

B

39.

39.

pit depth ~ ¼ λ



path diff = 2d = ½ λ

B

40.

40.

Increased quantum efficiency → more electrons per pixel → greater voltage → increased signal.

A

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